Lesson 7.3 Guided Notes

Session 1 · Absolute Value as Distance

I can use absolute value to describe how far a number is from zero. · 6.NS.C.7c

Word / Spanish	Meaning — complete the blank	Example
absolute value / valor absoluto	How many units a number is from _____.	|−3| = 3 and |3| = 3
distance / distancia	How many units apart two points are. It is never _____.	from 0 to −5 is 5 units
magnitude / magnitud	How big a number is, ignoring its _____.	−50 has magnitude 50

Key idea

Absolute value is never _____, because distance is never negative.

Closer to 0 means a _____ absolute value.

Worked example

Find |−13| and |6|. Which number is farther from 0?

−13 is _____ units left of 0, so |−13| = 13.

6 is 6 units right of 0, so |6| = _____.

Answer: 13 > 6, so _____ is farther from 0.

You try

1. Find each absolute value.
a. |−21| = _____
b. |4.5| = _____

_____

2. Which number is closer to 0: −2.5 or 3? Explain with absolute value.

_____

Sum it up: Absolute value is never negative because ___.

_____

Session 2 · Absolute Value as Total Distance

I can use absolute value to find the total distance between a negative and a positive reading. · 6.NS.C.7c

Word / Spanish	Meaning — complete the blank	Example
absolute value / valor absoluto	How many units a number is from _____.	|−50| = 50
distance / distancia	How many units _____ two points are.	from −8 to 3 is 11 units
magnitude / magnitud	The _____ of a number, ignoring its sign.	−50 and 50 have the same size

Key idea

If two numbers are on _____ sides of 0, add their absolute values.

If they are on the _____ side of 0, subtract the smaller absolute value from the larger.

Worked example

Each unit is 1 block. How far is it from −8 to 3?

0 is between them, so they are on _____ sides.

|−8| = 8 and |3| = _____.

Answer: 8 + 3 = _____ blocks

You try

1. Find the distance between each pair.
a. −5 and 9 _____
b. −12 and −4 _____

_____

2. A submarine rises from −30 m to −12 m. How far does it rise?

_____

Sum it up: To find the distance between a negative and a positive number, I ___ because ___.

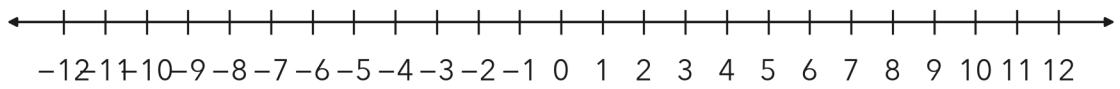
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Session 1 · Model, reason, and explain

Absolute Value as Distance · 6.NS.C.7c

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Compare −9, its opposite, and its absolute value. Which is greater as a number: −9 or 4? Which is farther from zero?



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_____

Extend your reasoning

Create a contrasting example that exposes the misconception: uses absolute value to decide ordinary numerical order.. Explain what changes and what stays true.

_____

_____

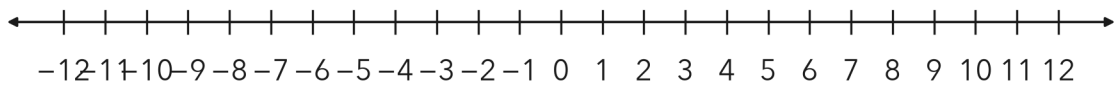
Self-check: Did my model, calculation, explanation, and units agree?

Session 2 · Model, reason, and explain

Absolute Value as Total Distance · 6.NS.C.7c

Success: represent the quantities, justify the relationship, and check the result with the correct units.

Find distance from −6 to 4 and from −6 to −2. Draw both segments. Explain why adding absolute values fits only the first case.



_____

_____

_____

Extend your reasoning

Create a contrasting example that exposes the misconception: always adds absolute values for any pair of numbers.. Explain what changes and what stays true.

_____

_____

Self-check: Did my model, calculation, explanation, and units agree?